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EXPLORING CALCULUS CONCEPTS WITH MAPLE.

Abstract:

In calculus, we study techniques for finding the maximum and minimum values of functions, as well as evaluating and visualizing volumes of solids of revolution formed by rotating regions bounded by graphs over given intervals about horizontal or vertical axes. These concepts will be explored through illustrative examples using **MAPLE.**

Discussion:

In our differential calculus classes, we explore the concepts of limits, continuity, and differentiability of real valued functions of a real variable. In my situation as an undergraduate four decades ago we were never introduced any technology to explore these concepts. The only way we were taught to explore these concepts was by using paper and pencil. In recent years we have used available technologies for effective teaching and learning of these concepts. Most calculus teachers are using either handheld calculators or software such as Maple or Mathematica. We are going to use Maple software to explore calculus concepts such as evaluating the absolute maximum and minimum of nicer functions or visualizing and computing the volume of revolution. We even explore the concept using Maple first and then introduce the paper and pencil afterwards depending on the complexity of the concept.

Maple Segments/Examples:

Example1: Find the maximum and minimum of the function $f(x) = x^4 - 4x^3 - 2x^2 + 12x - 7$ on the interval $[-2,4]$. Use Maple to plot the graph of the function and to visualize the critical points, the maximum and minimum points.

Solution Using Maple: (Example1-Code)

The Maple code below shows how to use Maple to find the critical points, then to evaluate the function at those critical points and the end points of the interval. Also we will compare these values to find the absolute maximum and the minimum of the function.

```
with(Student[Calculus1]):with(plots):with(student):
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> $f := x \to x^4 - 4x^3 - 2x^2 + 12x - 7;$

$$f := x \mapsto x^4 - 4 \cdot x^3 - 2 \cdot x^2 + 12 \cdot x - 7 \quad (1)$$

> $diff(f(x), x);$

$$4x^3 - 12x^2 - 4x + 12 \quad (2)$$

> $solve((2));$

$$1, 3, -1 \quad (3)$$

> $f(-2); f(-1); f(1); f(3); f(4);$

$$9$$
$$-16$$
$$0$$
$$-16$$
$$9 \quad (4)$$

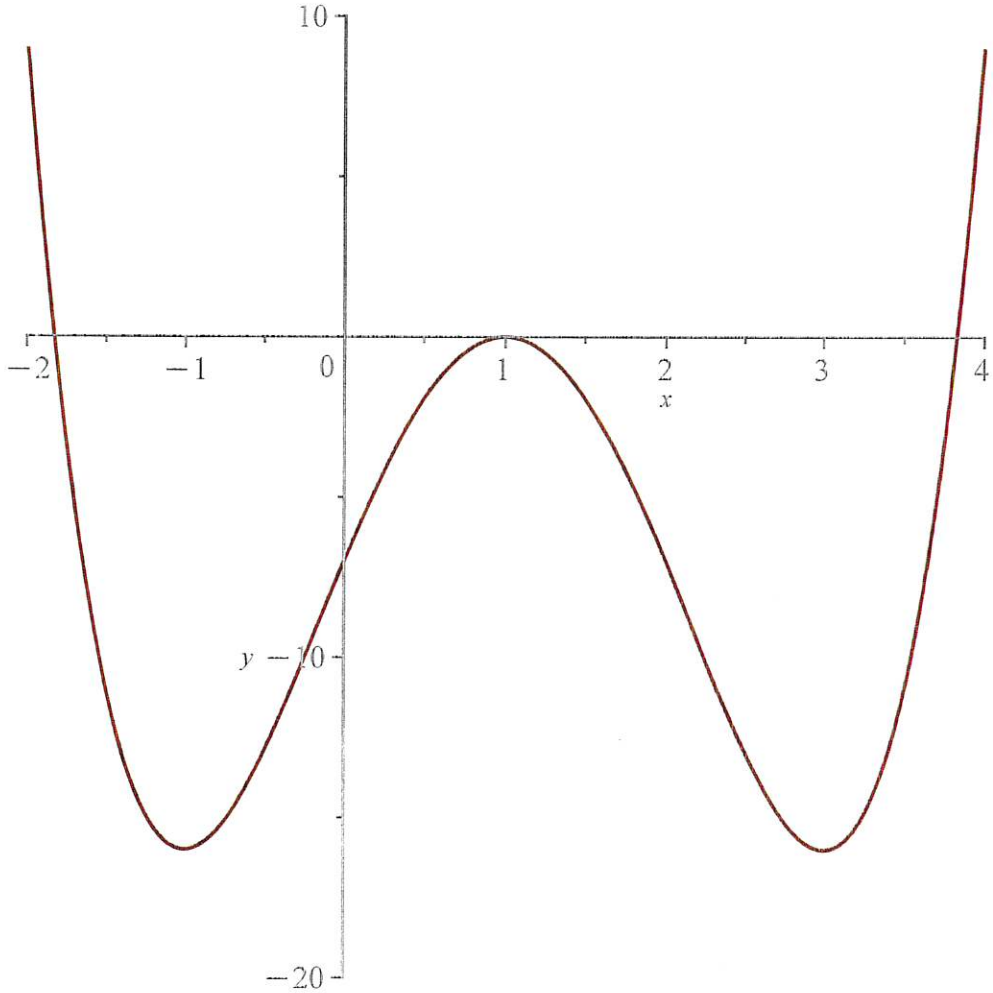
> $maximize(f(x), x = -2..4);$

$$9 \quad (5)$$

> $minimize(f(x), x = -2..4);$

$$-16 \quad (6)$$

> $plot(f(x), x = -2..4, y = -20..10);$


10
−2
−1
0
1
2
3
4
x
y
−10
−20

Example 2: Find the volume of the solid of revolution revolution of the region bounded by the graph of the function $f(x) = x^2, x = 0, x = 1$ and the x −axis rotated through 360^0about the $x - axis, y - axis$. Also show the volume of the solid of revolution by the use of MAPLE.

> $with(Student[Calculus1]) : with(plots) :$

> $f := x \rightarrow x^2;$

$$f := x \mapsto x^2 \tag{1}$$

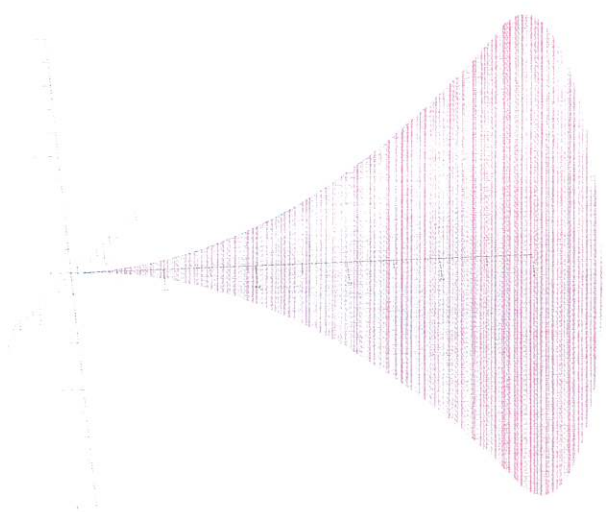
> $VolumeOfRevolution(f(x), x = 0..1, axis = horizontal, output = integral);$

$$\int_0^1 \pi x^4 \, dx \tag{2}$$

> $VolumeOfRevolution(f(x), x = 0..1, axis = horizontal, output = value);$

$$\frac{\pi}{5} \tag{3}$$

> $VolumeOfRevolution(f(x), x = 0..1, axis = horizontal, output = plot);$



The solid of revolution created on $0 \leq x \leq 1$ by rotation of $x^2 = x^2$ about the a

>

> $f := x \rightarrow x^2;$

$$f := x \mapsto x^2 \tag{4}$$

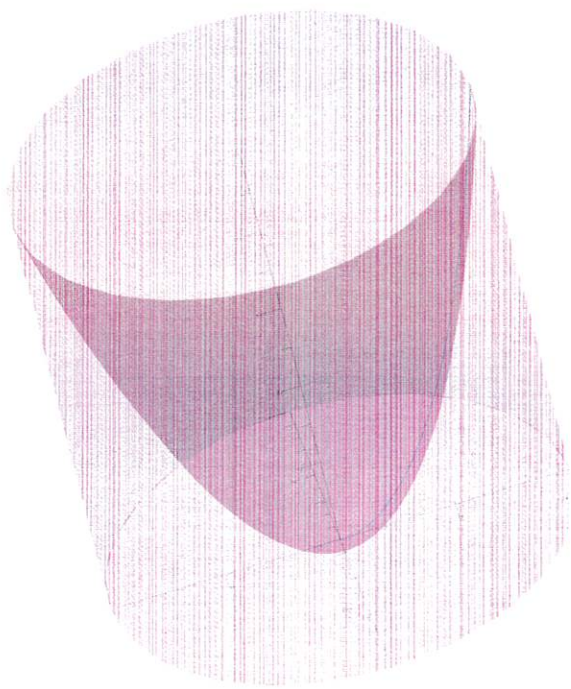
> $VolumeOfRevolution(f(x), x = 0..1, axis = vertical, output = integral);$

$$\int_0^1 2\,\pi\,x^3\,\mathrm{d}x \tag{5}$$

> $VolumeOfRevolution(f(x), x = 0..1, axis = vertical, output = value);$

$$\frac{\pi}{2} \tag{6}$$

> $VolumeOfRevolution(f(x), x = 0..1, axis = vertical, output = plot);$



The solid of revolution created on $0 \leq x \leq 1$ by rotation of $x^2 = x^2$ about the a

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References:

(1) DIFFERENTIAL CACULUS WITH MAPLE: Book published by LINUS BOOKS (2020), S. Velummylum, P. Varatharajah, K. Mahalingam: ISBN 13:978-1-60797-904-7

(2) Soma Velummylum: Use of Excel Spread Sheet to Approximate the Zero of a Differentiable Function. 18th Annual International Conference on Technology for Collegiate Mathematics , Hilton in the Walt Disney World Resort, Orlando, Florida, March 16-19-2006.